



케이녹 웹해킹 4회차 과제

17기 최동현 (202511688)

1-1. Git (Version Control System)

- Git(VCS, 버전 관리 시스템)이란?

: **Git**은 소스코드 파일의 **변경 사항**을 **추적**하기 위해 클라이언트 환경에서 동작하는 **분산 버전 관리 시스템**(DVCS, Distributed Version Control System)입니다.

델타(Delta, 변경된 부분)만을 저장하는 기존 시스템과 달리, Git은 파일 시스템의 상태를 **스냅샷**(Snapshot) **형태**(변경 시점 전체 상태 저장)로 캡처하고 **SHA-1 Hash** 알고리즘을 통해 데이터의 **무결성**을 **검증**합니다.

여담으로 Git을 만든 리누스 토르발즈가 Linux Kernel을 개발 하면서 버전 관리를 CVS, SVN으로 하다가 답답해서(빡쳐서) Git을 만들었다는 설이 있습니다.

*CVS and SVN are “**ugly and stupid**” and “have a serious **brain injury**.”
- Linus Torvalds*

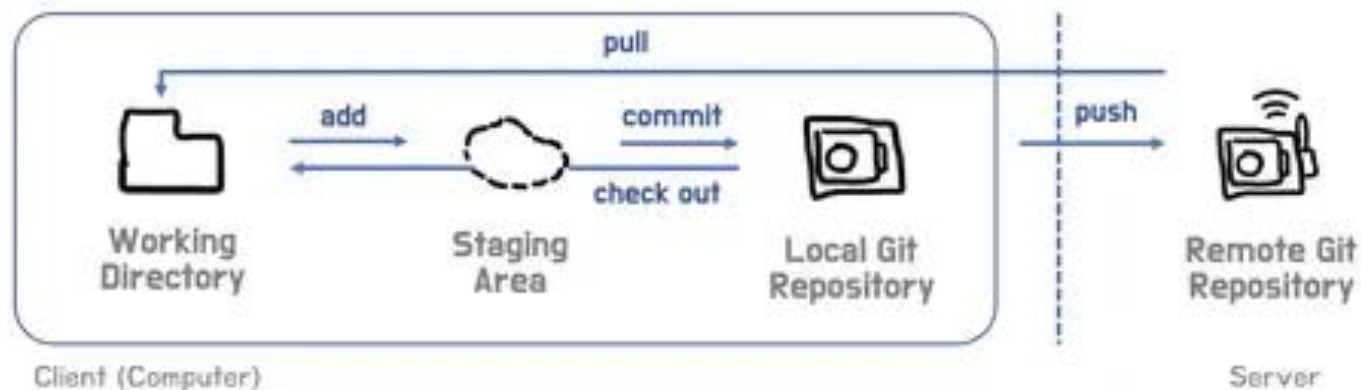


1-2. Git (Version Control System)

• Git의 구조

: Git의 로컬 환경은 물리적 위치에 따라 3가지 논리적 구역으로 나뉩니다.

- 1) **Working Directory (작업 디렉토리)**: 사용자가 현재 코드를 작성하고 수정하는 실제 파일 시스템 영역
- 2) **Staging Area (스테이징 영역)**: 다음 버전으로 기록(Commit)할 파일들의 메타데이터가 임시로 저장되는 가상의 대기열
- 3) **Local Repository (로컬 저장소 [.git])**: 스테이징 영역의 데이터가 스냅샷 형태로 영구적으로 기록(Commit)되는 로컬 데이터베이스



1-3. Git 실습 (#1)

```
35% 6% 13 GB
Apple > ~/Desktop/CHOI/TestRepo gh repo create TestRepo --private
✓ Created repository NullBins/TestRepo on github.com
https://github.com/NullBins/TestRepo

Apple > ~/Desktop/CHOI/TestRepo git init
Initialized empty Git repository in /Users/choi/Desktop/CHOI/TestRepo/.git/

Apple > ~/Desktop/CHOI/TestRepo > git main echo "# Hello Git" > ./README.md

Apple > ~/De/CHOI/TestRepo > git main ?1 git add README.md

Apple > ~/De/CHOI/TestRepo > git main +1 git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
    new file:   README.md
```

1-3. Git 실습 (#2)

```
32% 6% 13 GB 0.0 kB↓
> ~/Desktop/CHOI/TestRepo > git main +1 git commit -m "1st commit"
[main (root-commit) 19a0a34] 1st commit
1 file changed, 1 insertion(+)
create mode 100644 README.md

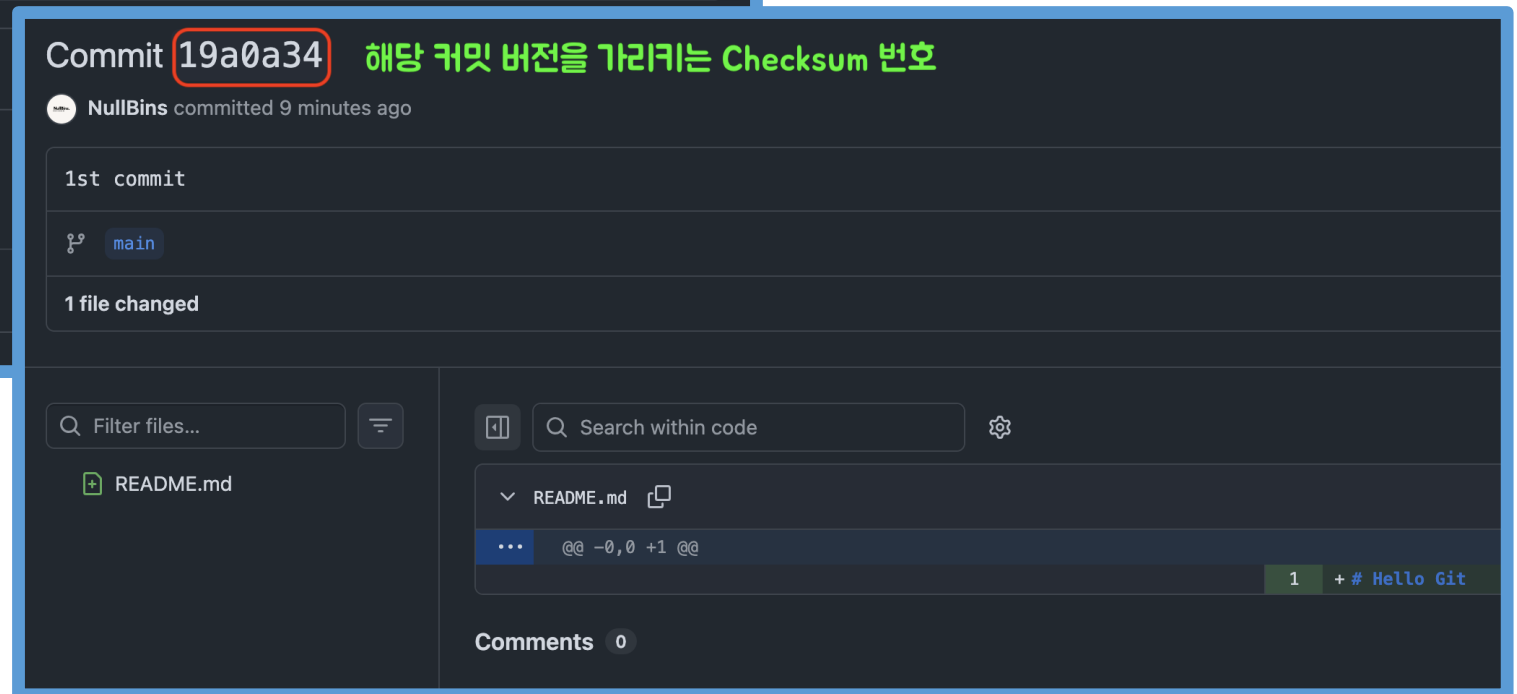
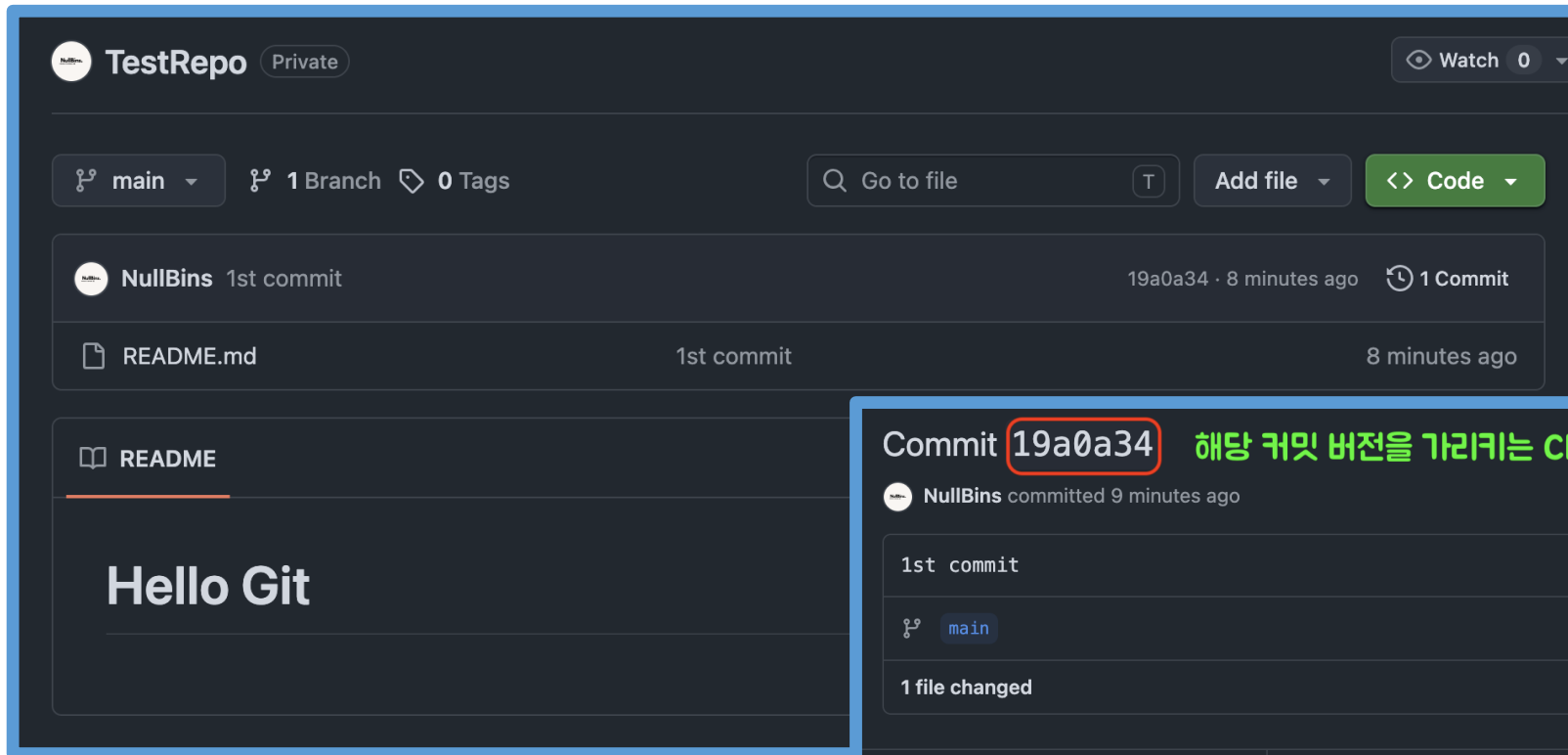
> ~/Desktop/CHOI/TestRepo > git main git branch -M main

> ~/Desktop/CHOI/TestRepo > git main git remote add origin https://github.com/NullBins/TestRepo.git

> ~/Desktop/CHOI/TestRepo > git main git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 225 bytes | 225.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/NullBins/TestRepo.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

> ~/Desktop/CHOI/TestRepo > git main
```

1-3. Git 실습 (#3)



2. GitHub (Remote Repository)

- 그러면 GitHub이란?

: **GitHub**는 Git의 로컬 저장소를 동기화하여 인터넷 상에 **백업**하고, 다수의 개발자가 코드를 **병합(Merge)**할 수 있도록 **원격 저장소(Remote Repository)**를 제공하는 **클라우드 호스팅 서비스**입니다.

이외에도 PR이나 Action 등 여러 많은 기능들과 소통할 수 있는 커뮤니티 기능들을 지원합니다.



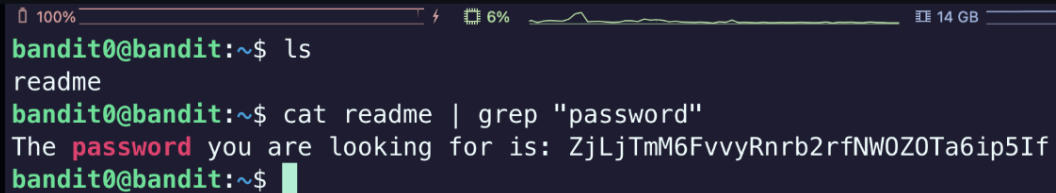
3. 실습 과제 풀이 (Bandit)

[Level0 시작]

- Username: bandit0
- PW: bandit0

1. Solve Commands

```
ls
cat readme
```



A terminal window showing the execution of commands to solve Level 0. The user runs 'ls' and 'cat readme'. The output of 'cat readme' shows a message: 'The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If'.

```
bandit0@bandit:~$ ls
readme
bandit0@bandit:~$ cat readme | grep "password"
The password you are looking for is: ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If
bandit0@bandit:~$
```

2. Flag

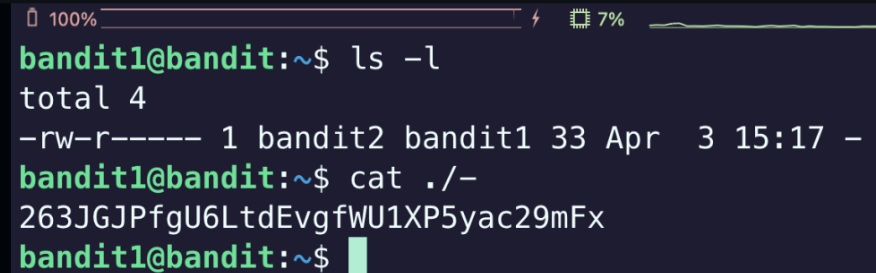
ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If

[Level1 풀이]

- Username: bandit1
- PW: ZjLjTmM6FvvyRnrb2rfNW0Z0Ta6ip5If

1. Solve Commands

```
ls -l
cat ./-
```



A terminal window showing the execution of commands to solve Level 1. The user runs 'ls -l' and 'cat ./-'. The output of 'cat ./-' shows a file named 'bandit2' with permissions '-rw-r-----' and owner 'bandit1'. The user then runs 'cat ./-' again to get the password.

```
bandit1@bandit:~$ ls -l
total 4
-rw-r----- 1 bandit2 bandit1 33 Apr  3 15:17 -
bandit1@bandit:~$ cat ./-
263JGJPfgU6LtdEvgfWU1XP5yac29mFx
bandit1@bandit:~$
```

2. Flag

263JGJPfgU6LtdEvgfWU1XP5yac29mFx

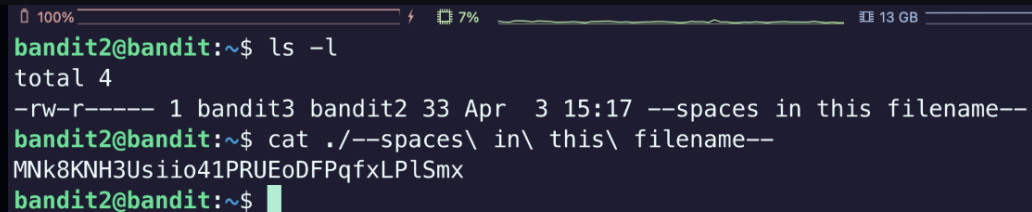
3. 실습 과제 풀이 (Bandit)

[Level2 풀이]

- Username: bandit2
- PW: 263JGJPfgU6LtdEvgfWU1XP5yac29mFx

1. Solve Commands

```
ls -l
cat ./--spaces\ in\ this\ filename--
```



```
bandit2@bandit:~$ ls -l
total 4
-rw-r----- 1 bandit3 bandit2 33 Apr  3 15:17 --spaces in this filename--
bandit2@bandit:~$ cat ./--spaces\ in\ this\ filename--
MNk8KNH3Usiio41PRUEoDFPqfxLPLSmx
bandit2@bandit:~$
```

2. Flag

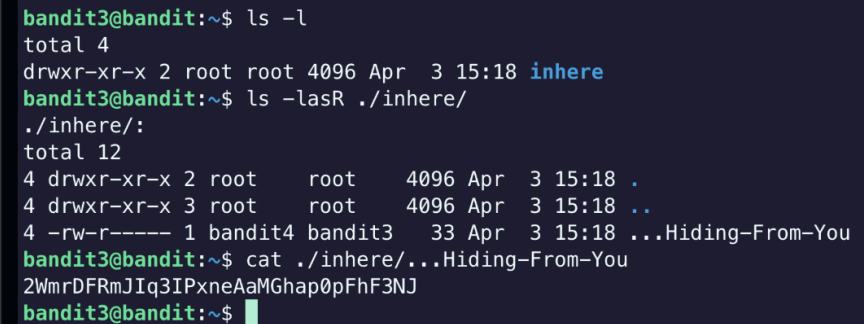
MNk8KNH3Usiio41PRUEoDFPqfxLPLSmx

[Level3 풀이]

- Username: bandit3
- PW: MNk8KNH3Usiio41PRUEoDFPqfxLPLSmx

1. Solve Commands

```
ls -l
ls -lasR ./inhere/
cat ./inhere/...Hiding-From-You
```



```
bandit3@bandit:~$ ls -l
total 4
drwxr-xr-x 2 root root 4096 Apr  3 15:18 inhere
bandit3@bandit:~$ ls -lasR ./inhere/
./inhere/:
total 12
4 drwxr-xr-x 2 root root 4096 Apr  3 15:18 .
4 drwxr-xr-x 3 root root 4096 Apr  3 15:18 ..
4 -rw-r----- 1 bandit4 bandit3 33 Apr  3 15:18 ...Hiding-From-You
bandit3@bandit:~$ cat ./inhere/...Hiding-From-You
2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ
bandit3@bandit:~$
```

2. Flag

2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ

3. 실습 과제 풀이 (Bandit)

[Level4 풀이]

- Username: bandit4
- PW: 2WmrDFRmJIq3IPxneAaMGhap0pFhF3NJ

1. Solve Commands

```
ls -l
ls -lasR ./inhere
find ./inhere -type f -exec cat -n {} \;
```

```
bandit4@bandit:~$ ls -l
total 4
drwxr-xr-x 2 root root 4096 Apr  3 15:18 inhere
bandit4@bandit:~$ ls -lasR ./inhere/
./inhere/:
total 48
4 drwxr-xr-x 2 root root 4096 Apr  3 15:18 .
4 drwxr-xr-x 3 root root 4096 Apr  3 15:18 ..
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file00
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file01
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file02
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file03
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file04
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file05
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file06
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file07
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file08
4 -rw-r----- 1 bandit5 bandit4 33 Apr  3 15:18 -file09
bandit4@bandit:~$ find ./inhere/ -type f -exec cat -n {} \;
1 00-00c000;Kde{f
+<0 1 4t0:0z0l0)000Lm0L0
Y016090000M 1 40QYVPkxZ00E005pTW81FB8j8lxXGUQw
1 00y0er'0v>/ a@.0'm000000 1 0-0v0000000hH0X00i>+0I0~0aP08Q 1 00P0vv0)0H0:0I0$0$0X0
2 0j0 1 VN0F00#00;0:t0000v0z00:0#0 1 00uB0(N0000;01-00s00$a001m 1 o"00 ,0
i0M0
-g0x,00v000z0 1 0300P0WD 00~^c@0000#Erttbbandit4@bandit:~$
```

2. Flag

4oQYVPkxZ00E005pTW81FB8j8lxXGUQw

[Level5 풀이]

- Username: bandit5
- PW: 4oQYVPkxZ00E005pTW81FB8j8lxXGUQw

Problem description

The password for the next level is stored in a file somewhere under the inhere directory and has all of the following properties:

human-readable
1033 bytes in size
not executable

1. Solve Commands

```
ls -l
ls -lasR ./inhere
find ./inhere -type f -size 1033c -exec cat -n {} +
```

```
bandit5@bandit:~$ ls -l
total 4
drwxr-x--- 22 root bandit5 4096 Apr  3 15:18 inhere
bandit5@bandit:~$ ls -lasR ./inhere/
total 88
4 drwxr-x--- 22 root bandit5 4096 Apr  3 15:18 .
4 drwxr-xr-x 3 root root 4096 Apr  3 15:18 ..
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere00
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere01
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere02
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere03
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere04
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere05
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere06
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere07
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere08
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere09
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere10
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere11
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere12
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere13
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere14
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere15
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere16
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere17
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere18
4 drwxr-x--- 2 root bandit5 4096 Apr  3 15:18 maybehere19
bandit5@bandit:~$ find ./inhere/ -type f -size 1033c -exec cat -n {} +
1 HWasnPhtq9AVKe0dmk45nxy20cvUa6EG
```

3. 실습 과제 풀이 (Bandit)

[Level6 풀이]

- Username: `bandit6`
- PW: `HWasnPhtq9AVKe0dmk45nxy20cvUa6EG`

Problem description

The password for the next level is stored somewhere on the server and has all of the following properties:

- owned by user bandit7
- owned by group bandit6
- 33 bytes in size

1. Solve Commands

```
ls -al
find / -type f -size 33c -exec stat -c "%n %U:%G" {} +
2>/dev/null | grep bandit7:bandit6
cat /var/lib/dpkg/info/bandit7.password
```

```
bandit6@bandit:~$ ls -al
total 20
drwxr-xr-x  2 root root 4096 Apr  3 15:17 .
drwxr-xr-x 150 root root 4096 Apr  3 15:20 ..
-rw-r--r--  1 root root  220 Mar 31  2024 .bash_logout
-rw-r--r--  1 root root 3851 Apr  3 15:10 .bashrc
-rw-r--r--  1 root root  807 Mar 31  2024 .profile
bandit6@bandit:~$ find / -type f -size 33c -exec stat -c "%n %U:%G" {} + 2>/dev/null | grep bandit7:bandit6
/var/lib/dpkg/info/bandit7.password bandit7:bandit6
bandit6@bandit:~$ cat /var/lib/dpkg/info/bandit7.password
morbNTDkSW6jIlUc0ym0dMaLn0lFVAaj
bandit6@bandit:~$
```

2. Flag

`morbNTDkSW6jIlUc0ym0dMaLn0lFVAaj`

[Level7 풀이]

- Username: `bandit7`
- PW: `morbNTDkSW6jIlUc0ym0dMaLn0lFVAaj`

Problem description

The password for the next level is stored in the file `data.txt` next to the word `millionth`

1. Solve Commands

```
ls -l
cat ./data.txt | grep millionth
```

```
bandit7@bandit:~$ ls -l
total 4088
-rw-r----- 1 bandit8 bandit7 4184396 Apr  3 15:18 data.txt
bandit7@bandit:~$ cat ./data.txt | grep millionth
millionth      dfwvzFQi4mU0wfNbF0e9RoWskMLg7eEc
bandit7@bandit:~$
```

2. Flag

`dfwvzFQi4mU0wfNbF0e9RoWskMLg7eEc`

3. 실습 과제 풀이 (Bandit)

[Level8 풀이]

- Username: bandit8
- PW: dfwvzFQi4mU0wfNbF0e9RoWskMLg7eEc

Problem description

The password for the next level is stored in the file data.txt and is the only line of text that occurs only once

1. Solve Commands

```
sort ./data.txt | uniq -u
```

```
bandit8@bandit:~$ ls -l
total 36
-rw-r----- 1 bandit9 bandit8 33033 Apr  3 15:18 data.txt
bandit8@bandit:~$ sort ./data.txt | uniq -u
4CKMh1JI91bUIZZPXDqGana14xvAg0JM
bandit8@bandit:~$
```

2. Flag

```
4CKMh1JI91bUIZZPXDqGana14xvAg0JM
```

[Level9 풀이]

- Username: bandit9
- PW: 4CKMh1JI91bUIZZPXDqGana14xvAg0JM

Problem description

The password for the next level is stored in the file data.txt in one of the few human-readable strings, preceded by several '=' characters.

1. Solve Commands

```
ls -l
strings ./data.txt | grep "="
```

```
bandit9@bandit:~$ ls -l
total 20
-rw-r----- 1 bandit10 bandit9 19382 Apr  3 15:17 data.txt
bandit9@bandit:~$ strings ./data.txt | grep "="
===== the
I\=0w
V?L=
%3=VZ
===== password
={M\
===== is
=Dvq
=n/N
===== FGUW5ilLVJrxX9kMYMm1N4MgbpfMiqey
zX]%=
]\{=
bandit9@bandit:~$
```

3. 실습 과제 풀이 (Bandit)

[Level10 풀이]

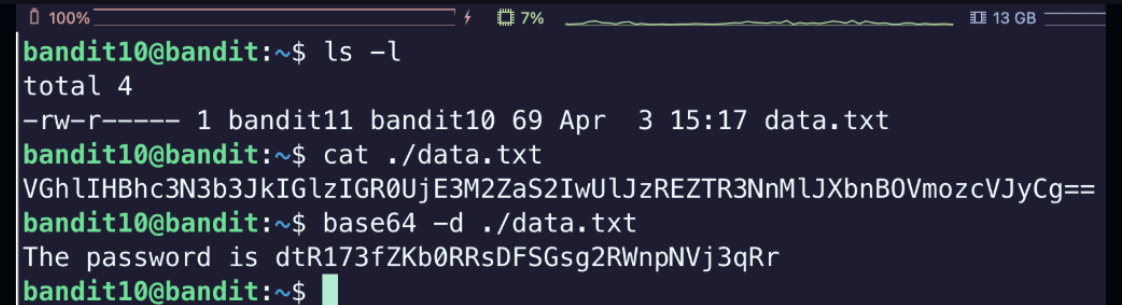
- Username: bandit10
- PW: FGUW5ilLVJrxX9kMYMmLN4MgbpfMikey

Problem description

The password for the next level is stored in the file data.txt, which contains base64 encoded data

1. Solve Commands

```
ls -l
cat ./data.txt
base64 -d ./data.txt
```



A terminal window showing the commands and their output. The terminal has a status bar at the top with 100% CPU, 7% battery, and 13 GB memory. The commands and output are as follows:

```
bandit10@bandit:~$ ls -l
total 4
-rw-r----- 1 bandit11 bandit10 69 Apr  3 15:17 data.txt
bandit10@bandit:~$ cat ./data.txt
VGhlIHBhc3N3b3JkIGlzIGR0UjE3M2ZaS2IwU1JzREZTR3NmMlJXbnB0VmozcVJyCg==
bandit10@bandit:~$ base64 -d ./data.txt
The password is dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr
bandit10@bandit:~$
```

2. Flag

dtR173fZKb0RRsDFSGsg2RWnpNVj3qRr



감사합니다

17기 최동현